Unit 2 Progress Check: FRQ Part B

This question involves processing strings of text.

Consider the following variables.

- colWidth is an int variable with a positive value.
- text is a String variable with a non-null value.
- otherText is a String variable with a non-null value.

1. A code segment, which you will write, should print output according to the following rules.

- If the length of text is greater than colWidth, then the value of text will be printed.
- Otherwise, the printed output will consist of colWidth characters contained between two "|" characters. The text between the "|" characters will consist of a sequence of periods ("."), followed by the value of text, followed by another sequence of periods.

  The printed output will have an equal number of periods on each side of text, if possible. If this is not possible, then there will be one fewer period on the left side of text than on the right side of text.

The intended behavior is illustrated in the following table.

Value of colWidth	Value of text	Printed Output
5	**"ab"**	|.ab..|
5	**"abc"**	|.abc.|
10	**"ab"**	|....ab....|
10	**"abc"**	|...abc....|
10	**"abcdefghij"**	|abcdefghij|
10	**"abcdefghijklmnopqrstuv"**	abcdefghijklmnopqrstuv

Write a code segment to print the intended output. Your implementation must meet all specifications and conform to the examples shown in the table.

2. A code segment, which you will write, should print the length of the longest prefix of the text that is a substring of otherText. A prefix is a substring that occurs at the beginning of another string.

The intended behavior is illustrated in the following table.

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Value of text	Value of otherText	Printed Output	Explanation
"apple"	"pear"	1	The prefix "a" is a substring of "pear", but "ap" is not.
"apple"	"pineapple"	5	The prefixes "a", "ap", "app", "appl", and "apple" are substrings of "pineapple". The length of the longest substring is 5.
"apple"	"happy"	3	The prefixes "a", "ap", and "app" are substrings of "happy". The length of the longest substring is 3.
"apple"	"leek"	0	No prefix of "apple" is a substring of "leek".

Write a code segment to print the intended output. Your implementation must meet all specifications and conform to the examples shown in the table.

This question involves writing two code segments that use information about downloaded songs.

A record company uses the following variables to maintain information about each song in the company's catalog.

- `downloads` is an `int` variable that represents the number of times a song was downloaded. Assume the value of this variable is positive.
- `songInfo` is a `String` variable that contains information about the song. The string consists of the artist name, followed by a hyphen ("-"), followed by the album title, followed by another hyphen, followed by the song title. Assume that no other hyphens appear in the string.

3. A code segment, which you will write, should determine whether the ones digit of the number stored in `downloads` appears exactly one other time in the number. If so, the code segment should print "exactly once"; otherwise, it should print "not exactly once".

For example, if `downloads` has a value of 1734567, then the segment should print "exactly once" because the ones digit, which is 7, appears one other time.

As another example, if `downloads` has a value of 325322, then the segment should print "not exactly once" because the ones digit, which is 2, appears two other times.

Write a code segment to print the intended output. Your implementation must conform to the examples.

4. A code segment, which you will write, should process the string `songInfo` and print the song title and artist name separated by the string (" by ").

For example, if `songInfo` is "Carrie Ferriero-Greatest Hits-Tell Me Why", then the code segment should print "Tell Me Why by Carrie Ferriero".

As another example, if `songInfo` is "Aniita-Mixtape 2-How Much More", then the code segment should print "How Much More by Aniita".

Write a code segment to print the intended output. Your implementation must conform to the examples.

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